Crewpto

Easy on friendships. Easier on your wallets.



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Executive Summary

Revolutionizing Group Expenses

Crewpto combines the simplicity of traditional expense-sharing apps with the power of blockchain technology. It offers users the ability to track and settle group expenses in real-time using cryptocurrencies, ensuring low-cost and efficient transactions.

Seamless Crypto Settlements

With a custom smart contract at its core, Crewpto simplifies debt management and enables direct peer-to-peer settlements. This eliminates the need for third-party payment platforms, making group transactions more efficient and borderless.

Secure and Transparent Platform

Integrated with Metamask wallets, Crewpto prioritizes user security through private key encryption. Every transaction is recorded on the blockchain, offering transparency and trust while keeping the process user-friendly.

Problem Statement

Managing shared expenses with traditional platforms like Splitwise is inefficient and costly, especially for cross-border transactions, as they rely on external tools for settlements.

Cryptocurrencies are gaining strong traction as a secure and transparent medium.

There is a need for a convenient, cost effective and efficient platform for cross-border transaction that integrates expense tracking with seamless crypto-based settlements.





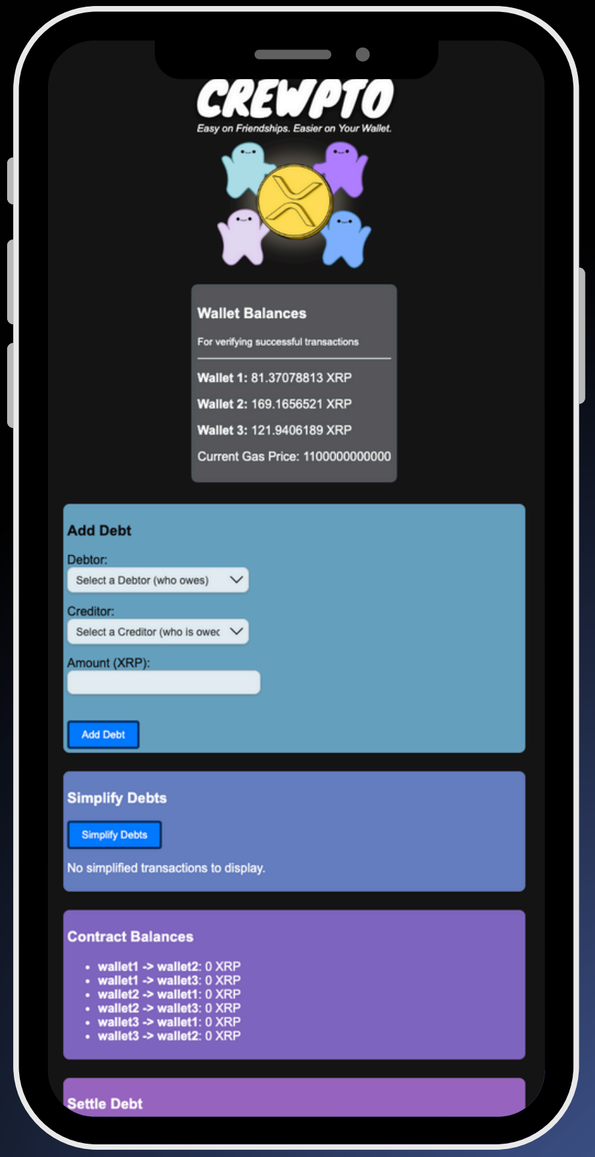


Our Solution: Crewpto

Cryptocurrency exchanges in a few simple steps

Crewpto is a blockchain platform for group expense management, integrating Metamask, smart contracts, and the XRPL EVM Sidechain Testnet. It combines real-time tracking with seamless crypto settlements in a decentralized ecosystem.

Metamask ensures secure wallet connectivity, while smart contracts handle debt tracking, balance simplification, and direct payments. The XRPL EVM Sidechain offers scalable, low-cost, and Ethereum-compatible transactions for efficient settlements.



General Tech Stack

Frontend HTML UI

- Designed for user interaction with Flask templates to dynamically display wallet and contract balances.
- Features forms for adding debts, simplifying transactions, and settling payments.

Backend Flask Application

- Built with Python and integrated with the blockchain using Web3.py.
- Exposes key APIs for adding, simplifying, and settling debts.

Smart Contract

- Leverages the XRPL EVM Sidechain for scalability and low-cost blockchain transactions.
- Records debts securely on-chain.
- Reduces transactions by calculating net balances.
- Facilitates direct payments in XRP between wallets.

Remix IDE and Metamask

- MetaMask provides secure connectivity for signing and validating transactions on-chain.
- Remix provides an environment for smart contract testing and deployment, supplying the contract address and ABI for seamless full stack integration.

User
↓
Frontend (HTML/Flask)
↓
Routes (/add_debt, /simplify, /send)
↓
Backend (Flask + Web3.py)
↓
MetaMask (Wallet Authorization)
↓
Blockchain (XRPL EVM Sidechain)
↓
Smart Contract (new_splitwise.sol)
↓
Updates Balances / Settles Payments
↓
Feedback to Backend
↓
Frontend (Display Results)
↓
User

Competitor Analysis

Crewpto’s Unique Value Proposition

- Blockchain-powered, ensuring security, transparency, and low costs.
- Designed for group expense management, filling a unique gap in the market.
- Scalability with XRPL EVM Sidechain
- Simplify feature reduces the number of transactions made, further lowering costs.

1. Traditional Expense Management Platforms (e.g., Splitwise)

- Widely adopted for group expense tracking but does not have integrated settlement option; users rely on external payment platforms (e.g., PayPal, Venmo)

2. Cryptocurrency Payment Platforms (e.g., BitPay, Coinbase Commerce)

- Focused on e-commerce and business use rather than peer-to-peer transactions.
- Not designed for group expense management.

3. Decentralized Finance (DeFi) Platforms (e.g., Aave, Compound

- Advanced use of smart contracts for transactions but complex user experience unsuitable for casual users

4. Peer-to-Peer (P2P) Payment Platforms (e.g., Venmo, PayPal)

- High fees for international and cross-currency transactions and no focus on group expense management